

Ethan Joseph

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EDUCATION

Bachelor of Engineering (B. Eng) – Software Engineering

McMaster University

Graduation Date: June 2030

PROJECTS

Sentinel Security Auditor – [View Live Site](#) | Python, TypeScript, Gemini API, GitHub Actions, Vercel Nov 2025 – Mar 2026

- Built an AI-driven application security auditor that analyzes local and GitHub repositories to generate structured vulnerability findings, attack-flow graphs, and remediation guidance.
- Designed a repository ingestion and batching pipeline for large codebases, enabling analysis of projects **exceeding 50k LOC** under strict file size and context limits.
- Developed repository-aware vulnerability analysis that **reduced false positives by ~35%** vs rule-only scanners (benchmarked across 40+ open-source repos).
- Implemented automated remediation with structured Git patch generation, achieving **~70% successful fix** application rate as measured by post-scan verification and rescanning of patched code.
- Optimized prompt architecture and context compression, **reducing token usage ~28%** while maintaining detection accuracy.
- Deployed a production web interface on Vercel, averaging **~800 monthly visits** from users scanning their repositories for security vulnerabilities.

FallGuard - MacHacks Winner: [Best use of Gemini API](#) | JavaScript, React, Tailwind CSS, Flask, Gemini API March 2026

- Built and deployed a browser-based AI application that analyzes images of environments to detect safety hazards and generate structured, actionable risk reports
- Engineered a Gemini-powered vision pipeline with normalized JSON outputs, supporting multi-image analysis (**up to 8 images** per request) with consistent structured responses
- Designed a weighted safety scoring system (0–100) linking hazard severity to prioritized recommendations, improving clarity of risk mitigation decisions
- Developed a full-stack architecture (vanilla JS + Tailwind frontend, Vercel serverless backend) handling client-side image compression and API processing, **reducing average payload size by ~60%** before inference
- Implemented context-aware follow-up analysis and local history persistence (25-entry cap), enabling faster re-analysis without re-uploading images.
- Achieved near real-time performance (**~2–4s per request** under typical usage) through optimized request handling and prompt design

WORK EXPERIENCE

Founder & Organizer - Eclipse Hackathons - [View Devpost](#) August 2024 – November 2024

- Founded and led a multi-school hackathon addressing globally relevant challenges, attracting **80+ participants** from 15+ schools across Ontario.
- Designed and developed the complete digital infrastructure for the event, including the public-facing website, registration system, submission portal, automated email workflows, and sponsor integrations, supporting 80+ registered users with zero downtime during live judging. Engineered backend workflows for application review and project uploads, reducing manual coordination overhead by **~60%** compared to form-based event systems.
- Secured corporate sponsorship from **HCLTech (~\$1000 value)** and recruited industry judges from AMD, Dropbox, Fractal Analytics, and SAP Concur, coordinating structured scoring across 20+ project submissions.
- Led end-to-end marketing strategy (social media outreach, targeted school campaigns, email distribution), generating 3,000+ website visits and **converting 4% of visitors** into registered participants.

TECHNICAL SKILLS

Languages | Python, C, C++, TypeScript, Java, SQL, Arduino (Embedded C/C++)

Software Architecture | Data Structures & Algorithms, Object-Oriented Programming, REST API Development, Unit Testing, Debugging, Prompt Engineering, Context Optimization

Systems & Infrastructure | Git, GitHub Actions (CI/CD), Linux, CLI Development, Cloud Deployment (Vercel), Docker, LLM API Integration (Gemini)